

Poisson Regression, and the GLM Unification

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Abstract

The conditional-distribution recipe of the logistic note, applied to count data: $Y|X=x \sim \text{Poisson}(\lambda(x))$ with log-linear rate. The per-sample gradient is once again *residual times features* — and this third instance is no coincidence: for any exponential-family response with its canonical link, the gradient of the conditional NLL is $(\hat{\mu} - y)x$. That unification — generalized linear models — is derived here, making the linear, logistic, and Poisson notes three rows of one table.

1 The model

Counts $y \in \{0, 1, 2, \dots\}$ with

$$\Pr(Y = y \mid X = x) = e^{-\lambda(x)} \frac{\lambda(x)^y}{y!}, \quad \lambda(x) = e^{w^\top x} > 0, \quad (1)$$

the exponential (log link) keeping the rate positive exactly as the sigmoid kept a probability in $(0, 1)$. Poisson facts: $\mathbb{E}[Y|x] = \text{Var}(Y|x) = \lambda(x)$ (mean equals variance — visible in the script's first panel), and the log-linear form means each unit increase of a feature *multiplies* the expected count by e^{w_j} .

2 Conditional MLE

The per-sample NLL, dropping the $\log y_i!$ term (constant in w):

$$\mathcal{L}_i(w) = \lambda_i - y_i \log \lambda_i = e^{w^\top x_i} - y_i w^\top x_i. \quad (2)$$

Differentiate: $\nabla_w \mathcal{L}_i = e^{w^\top x_i} x_i - y_i x_i$, hence

$$\nabla_w \mathcal{L}_i = (\lambda_i - y_i) x_i. \quad (3)$$

Residual times features, again. Stationarity $\sum_i e^{w^\top x_i} x_i = \sum_i y_i x_i$ is transcendental — no closed form, as in the logistic case — but the Hessian

$$\nabla^2 \mathcal{L} = \frac{1}{n} X^\top \text{diag}(\lambda_i) X \succeq 0 \quad (4)$$

is PSD with weights equal to the conditional *variances* (for Poisson, the rates themselves), so the loss is convex and Newton/IRLS — each step a weighted least-squares solve — converges quadratically (8 steps vs GD's thousands in the script).

3 The GLM unification

The pattern is now unmistakable:

response $Y x$	link $(\theta = w^\top x)$	mean $\hat{\mu}(\theta)$	per-sample gradient
Gaussian	identity	θ	$(\theta - y) x$
Bernoulli	logit	$\sigma(\theta)$	$(\sigma(\theta) - y) x$
Poisson	log	e^θ	$(e^\theta - y) x$

Proposition 1 (Canonical-link GLM gradient). *Let the conditional distribution be an exponential family in natural parameter θ , $p(y | \theta) = h(y) \exp(\theta y - A(\theta))$, with $\theta = w^\top x$ (the canonical link). Then the per-sample NLL is $\mathcal{L}_i(w) = A(w^\top x_i) - y_i w^\top x_i$ and*

$$\boxed{\nabla_w \mathcal{L}_i = (\hat{\mu}(w^\top x_i) - y_i) x_i, \quad \hat{\mu} = A' = \mathbb{E}[Y | \theta].} \quad (5)$$

Proof. $\nabla_w \mathcal{L}_i = A'(w^\top x_i) x_i - y_i x_i$, and the exponential-family identity $A'(\theta) = \mathbb{E}[Y | \theta]$ (differentiate $\int h e^{\theta y - A} = 1$ in θ) identifies A' as the mean function. Convexity is automatic: $A''(\theta) = \text{Var}(Y | \theta) \geq 0$, so the Hessian is $\frac{1}{n} X^\top \text{diag}(\text{Var}_i) X \succeq 0$ — which is also why the IRLS weights are always the conditional variances. $\square$

Check the table against the identity: Gaussian, $A = \theta^2/2$, $A' = \theta$; Bernoulli, $A = \log(1 + e^\theta)$, $A' = \sigma$; Poisson, $A = e^\theta$, $A' = e^\theta$. One theorem, all three gradients — and the reason “prediction minus target, pushed along the features” keeps reappearing: it is not a coincidence of examples but the derivative of the exponential family’s log-normalizer. Any new response distribution in this family (exponential, gamma, multinomial — see the softmax note) joins the table for free.

4 Experiment

$n = 5000$, $w^* = (0.3, 0.8, -0.5)$, $x \sim \mathcal{N}(0, I_2)$, counts drawn from (1) (seed 7). Newton/IRLS (8 steps) and GD (4000 steps) agree on the optimum, recovering w^* to sampling accuracy; hand and autograd SGD trajectories coincide to 10^{-10} ; the calibration panel plots fitted $e^{\hat{w}^\top x}$ against true $\lambda(x)$ on fresh points — the estimated object is the *rate function*, the Poisson sibling of logistic regression’s $p(x)$.